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# MLADAD: Machine Learning Algorithms Analysis on DDoS Attack Detection in SDN-IoT Networks

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## Abstract

Now a days, increase in applications of SDN-IoT networks for their effective management of network capabilities. Controller is the brain of the SDN-IoT networks and have the capabilities of managing and controlling the entire network. Due this feature controller become the main threat to the attackers and become the main vulnerable point is SDN. If an attacker is able to gain control of the controller, then the attacker may be able to control the whole SDN-IoT network. Its required to protect the controller and entire network from the attacker. In this regard, DDoS attack detection is considered in SDN-IoT networks is necessary. Machine learning algorithms such as Logistic regression, Decision Tree, and Random Forest with the intrusion dataset CSE-CIC-IDS2018 has been used. The results suggest that Random Forest and Decision Tree has best accuracy with respect to detection rate.

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**Keywords:** Software Defined Networks (SDN); machine Learning Techniques; Distributed Denial of Service (DDoS); Decision Tree; Logistic Regression; Random Forest

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## 1. Introduction

The industry is upgrading from Industry 4.0 to Industry 5.0 standard. The transformation happening by implementing more automation in the industry. This is can provided by introducing robotics and IOT sensor devices to perform the tasks and reduce human force. When, the automation increases the communication between these devices, need to happen smoothly and flawlessly [1]. This can be achieved through a programmable network architecture namely SDN-IOT architecture.

Software Defined Networking (SDN) architecture controls and manages the network with software rather than manual configurations on hardware, this offers flexibility, scalability, and programability [2]. SDN-IoT networking is building, managing and controlling the IoT networks that uses SDN architecture to simplify network management

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and improve flexibility and scalability. Due to demand for rapidly increasing new IoT devices there is also increasing in security risks. IoT devices are internet based and it consist of exclusive information leads safety issues. This leads various researchers are started exploring the different techniques to enhance the protection among these kinds of devices [3].

The global network view of the SDN controller allows it to filter network traffic and identify fraudulent movements. Although the separation of the control and data planes had many benefits, presented a new challenge in terms of being vulnerable to DDoS attacks [4]. In SDN networks, controllers are considered one of the most significant points of failure due to their high susceptibility to security breaches. Distributed Denial of Service (DDoS) attacks are a type of malicious cyberattack that can affect controllers in two ways: (i) directly, by flooding the controller with too many erroneous packets that interfere with its functionality; and (ii) indirectly, by attacking network nodes and causing the controller to be overloaded with control packets due to the SDN default forwarding policies configured in the switches [5]. The SDN network's control plane, which houses the controller 'the brain' and manages the whole network. This makes the network more susceptible. Attackers will find it more appealing to target if it fails, as a single point failure (SOF) affects the entire network [6].

There are various types of attacks on IoT network, one of them is DDoS attack [7]. In literature [8], there are lot studies are introduced to detect DDoS in SDN-IoT. The contribution of the paper as follows:

1. Detection of DDoS attacks in SDN-IoT Networks
2. Used different machine learning techniques, including Logistic regression, Decision Tree, and Random Forest

Remaining paper explains as follows: literature review with respect to overview of different attacks and DDoS attack detection techniques is presented in Section 2. Section 3 presents proposed system architecture. Dataset & pre-processing details in Section 4. Section 5 detailed about different classification models and the comparison between them with respect to performance evaluation followed by conclusion.

## 2. Literature Review

This Section will give the existing DDoS Detection technique followed by overview of different attacks present in SDN-IoT.

### 2.1. SDN IoT attacks

An attack on a network is a malicious activity that aims to disrupt, gain unauthorized access, or compromise the resources or infrastructure of a network. Network attacks can come in various forms and can target any plane of the SDN network [9]. Attack on SDN IoT Networks can include similar to those in SDN networks and additional considerations for the characteristics of IoT devices.

Types of attacks that can be targeted at SDN based IoT network include:

- **Device Spoofing:** Attackers Spoof their identities, MAC addresses or other network attributes to acquire unapproved entry to the SDN-based IoT network and manipulate or disrupt the functions of devices or the network [9].
- **Unauthorized Access:** Attackers may try access unauthorized to SDN principles or technologies or resources such as the SDN controller, switches or other network devices in order to manipulate the operation of the network [9].
- **Data Tampering:** Attackers can try to gain access to the information that is being shared and transmitted between the IoT devices and the SDN controller, leading to confidentiality breaches or manipulation of the behavior of the IoT device [9].
- **DDoS Attack:** DDoS (Distributed Denial of Service) is used to flood the enormous volume of requests or traffic on an SDN-IoT network, overloading the network resources, and causing disruption to normal operations [9].

Recognizing and thwarting DDoS assaults on servers and IoT networks, a novel LEDEM approach is implemented in [10]. This approach leverages cloud-SDN architecture in all aspects, including flexibility, dynamism, and programmability. Literature [11] presented a framework for DDOS attack detection on SDN IOT networks. The framework consists of a layer that calculates the length of the vectors by taking the cosine similarity of the packets. The quickly detects the assault using this strategy. However, it does not analyze machine learning techniques or false

positive procedures. Traffic-flow patterns were used by Xie et al. [12] to detect DDoS assaults. When compared to previous methods, our system demonstrates effective DDoS information detection with a relatively minimal overhead. But in situations when network traffic is high, this approach is insufficient, and a more advanced security solution must be put in place. A security system presented in [13] using unique machine learning (ML) approach that adapts automatically and increases the security on the Internet of Things (IoT).

FMDADM [14], SDN based system for IoT network DDoS attack detection and mitigation is proposed, which uses five tier design and four primary modules with a 32 packet window size. The first module provides an early detection based on the average drop rate. In second module, a unique double-check mapping function is used to help detect attacks at the data plane. The third module is a machine learning-based detection technique consists of four stages namely feature extraction, training and testing, classification, and data preparation. An attack mitigation procedure is used in the final module.

Using an SDN-WISE controller, the traffic flow-based DDoS attack detection method in [15] employs classifiers to identify assaults as soon as feasible; however, processing is done at the controller side, which results in a very small latency increase. A machine learning-based low-rate DDoS assault detection system was presented by the authors [16]. It employs stateful and stateless features from the Openflow package to identify attack traffic in the data plane and control plane of the SDN. The DDoS attacks on SDN network are shown in Table 1.

Table 1: DDOS Attacks in SDN-IOT networks - Literature Review

| S. No. | Paper                                                                                                     | Description                                                                     | Advantages                                                                                                            | Disadvantages                                                                       |
|--------|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 1.     | Learning-Driven Detection and Mitigation of DDoS Attack in IoT via SDN-Cloud Architecture [10]            | A learning-driven detection mitigation (LEDEM) detect and mitigate DDoS attacks | Uses the traffic flow in the network to identifies the attack and provide approximation approach to mitigate the flow | Not included other type of security breaches                                        |
| 2.     | A DDoS Attack Detection and Mitigation With Software-Defined Internet of Things Framework [11]            | The cosine similarity to check the DDos attacks in SDN framework                | This identifies the change in packet-in rate within short time depends on the threshold value                         | Single parameter is not enough to identify the attack more intelligence is required |
| 3.     | A Machine Learning Security Framework for IoT Systems [13]                                                | An AI framework uses ML to detect anomaly-based intrusions in IoT               | The anomalies detected with good accuracy                                                                             | The other metrics also need to provide strong solution                              |
| 4.     | Detection and Mitigation of DoS and DDoS Attacks in IoT-Based Stateful SDN: An Experimental Approach [17] | Detecting and countering DDoS attacks in IoT-SDN stateful environments          | The statistical solution to mitigate and detection of attacks                                                         | The more intelligence required for self configuration of parameters                 |
| 5.     | FMDADM [14]                                                                                               | The ML based detection algorithm used                                           | This framework defends both local and remote IoT nodes and to detects DDoS                                            | The processing time is high and also other types of attacks need to consider        |
| 5.     | Machine learning based low-rate DDoS attack detection for SDN enabled IoT networks [16]                   | A machine learning approach to detect                                           | More accurately identifying the attacks with minimum overhead                                                         | Other type of attacks need to consider                                              |
| 6.     | Machine-learning-assisted DDoS attack detection with P4 language [18]                                     | Machine learning with P4 for detecting DDoS attacks                             | More accurate in detection                                                                                            | Need to consider other attacks and also consider resource consumption of switch     |

### 3. Proposed System Architecture and Details

There are several steps involved in building a classification model of proposed system which is showed in Figure 1.

Dataset selection is the first and crucial step of model building, hence it requires identifying relevant datasets. Effective dataset selection ensures good analysis, modeling, or decision-making, which leads to good and accurate results.

Data pre-processing is a critical step in machine learning pipeline which involves analysis of the data, feature extraction, feature selection, visualization and challenges such as dealing with missing values, outliers, and inconsistencies in the data. Effective data possessing ensures high quality data which leads to better model performance. Scaling is a pre-processing technique used in machine learning to normalize or standardize the features of the dataset, which involves transforming the values of input features to common values so that they all have similar magnitudes which helps improving the performance. In this case, the given dataset is suitable for building a classification model which involves the process of selecting a classification, training the model on the train dataset, and evaluating its performance test dataset. As this is one of the crucial steps, it involves taking various steps into consideration such as model complexity, performance metrics, etc.

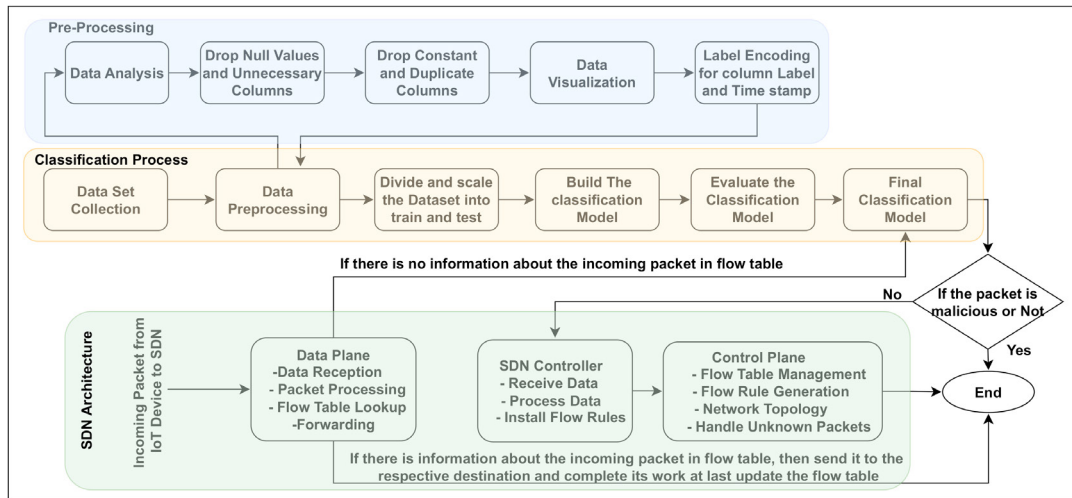


Fig. 1: Proposed Architecture

The data generated in IoT-SDN Network is collected at data plane is forwarded to classification model for checking malicious or not. If the data is malicious, proposed system is going to discard it. if it is normal data then only it will be forwarded to network. Same is shown in Figure 1 that will help build a classification model on the DDoS dataset and evaluate the performance of the model.

#### 4. Dataset Details

A joint study between the Canadian Institute for Cybersecurity (CIC) and the Communications Security Establishment (CSE) produced the dataset named CSE-CIC-IDS2018 [19] is used for the DDoS Detection in SDN-IoT Network. One of the main reason to choose this dataset is the latest one and has the features that are helpful for SDN-IoT networks compared to other datasets [20].

The size of this dataset is 6.89 GB. This dataset is itself a mixture of various datasets named after specific dates. Each subset gives information about different intrusion method possible. There are about ten sub-datasets. All sub-datasets come under classification.

One of the sub-dataset named 02-20-2018.csv has 84 columns and 02-21-2018.csv has 80 columns so by removing the extra 4 columns from 02-20-2018.csv and combining both the sub-data sets (Note: Those four columns were removed because they are classified as Repeated columns where they play no role while building model).

##### 4.1. System Details

The entire building and evaluation of the classification model is done using python programming language(Latest version of python was used - 3.12 version). The models were built on Notebooks that have 4 CPU cores and 30 GB of RAM, and GPU memory of 16GB. Numpy, Pandas, Seaborn, Matplotlib, sklearn Python libraries are used for visualizing the data and applying the appropriate algorithms on the data to build a classification model.

##### 4.2. Dataset Collection and pre-processing

From dataset CSE-CIC-IDS2018 [19], taking two sub datasets namely 02-20-2018.csv and 02-21-2018.csv and combine them to focus on DDoS Attacks detection. It contains 8997322 entries and 4 labels with number of rows as shown below:

- Benign (7733390) - Normal
- DDOS attack-LOIC-UDP (1730) - Malicious Data
- DDOS attacks-LOIC-HTTP (576191) - Malicious Data
- DDOS attack-HOIC (686012) - Malicious Data

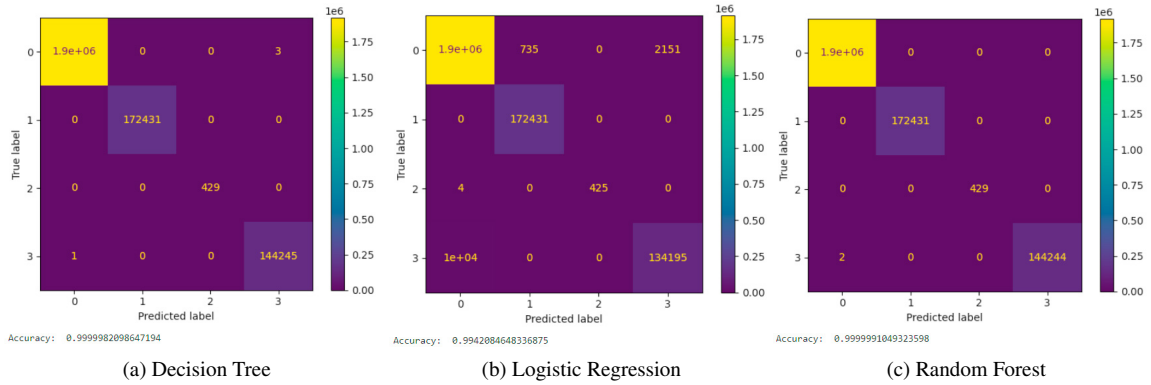


Fig. 2: Confusion Matrix 75% Training Data- 25% Testing Data

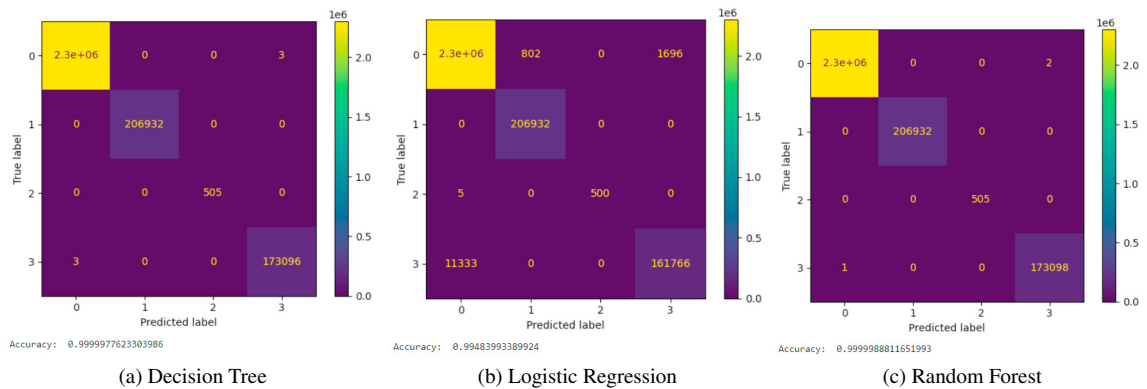


Fig. 3: Confusion Matrix 70% Training Data- 30% Testing Data

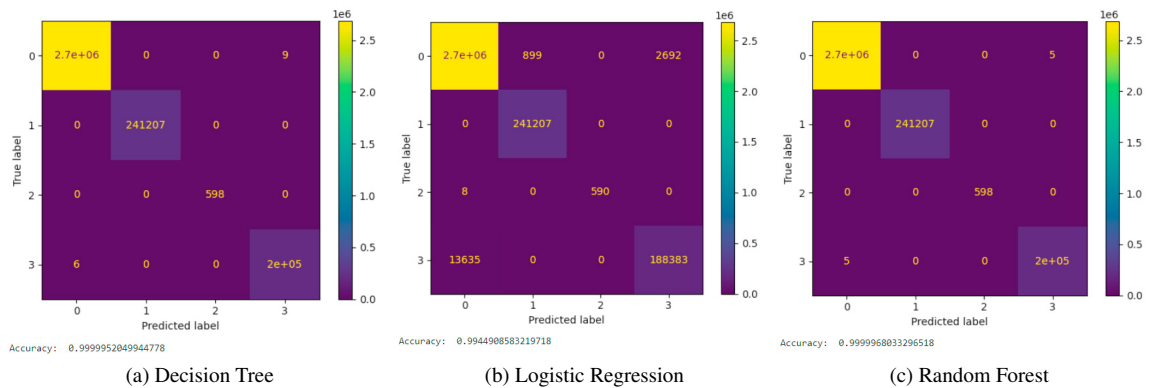
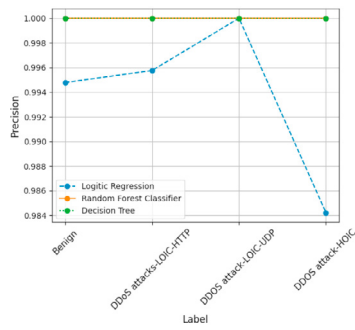
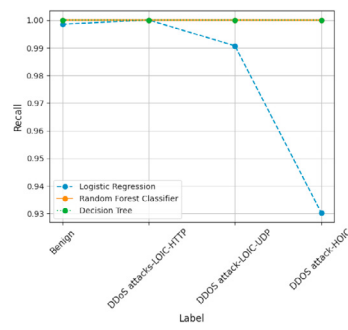


Fig. 4: Confusion Matrix 65% Training Data- 35% Testing Data

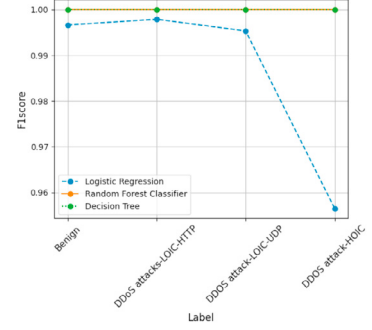
Coming to the Data pre-processing unnecessary columns and repeated columns will be removed because they contribute nothing while building the model. Unnecessary columns can be found by visualizing the correlation matrix using heatmap (a 2D visualization technique used for Geographical Representation of Data where colors are used to represent values).



(a) precision

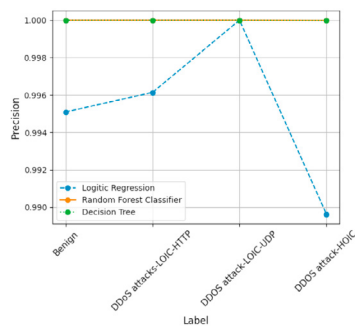


(b) Recall

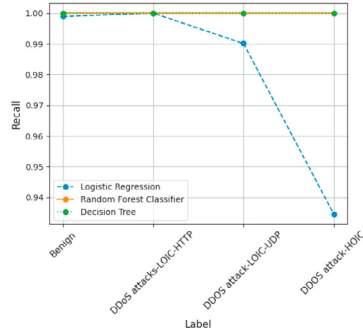


(c) F1 Score

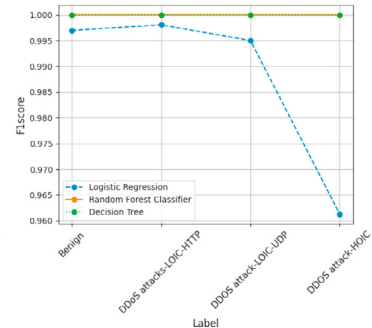
Fig. 5: 75% Training Data- 25% Testing Data



(a) precision

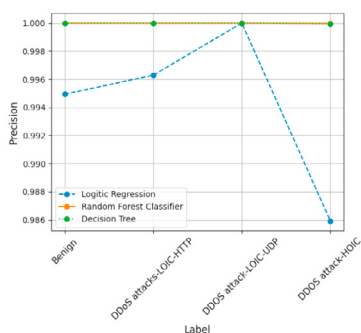


(b) Recall

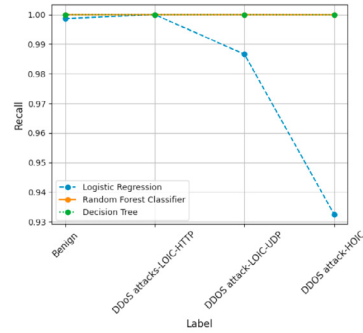


(c) F1 Score

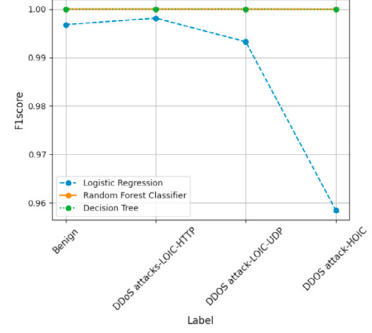
Fig. 6: 70% Training Data- 30% Testing Data



(a) precision



(b) Recall



(c) F1 Score

Fig. 7: 65% Training Data- 35% Testing Data

The label column determines whether the given data is related to DDoS attacks or not. So encoding should be done for Label Column in order to convert that to int datatype. The encoder that is used is Label Encoder. As data pre-processing involves Dealing with null values But there are no Null Values in the dataset. So, Next is splitting the dataset into train and test datasets. To do that, used different split sizes in the form train and test data 75-25, 70-30, 65-35 respectively and start using various classification algorithms. The test dataset contains only a single row named as Label and has four unique values.

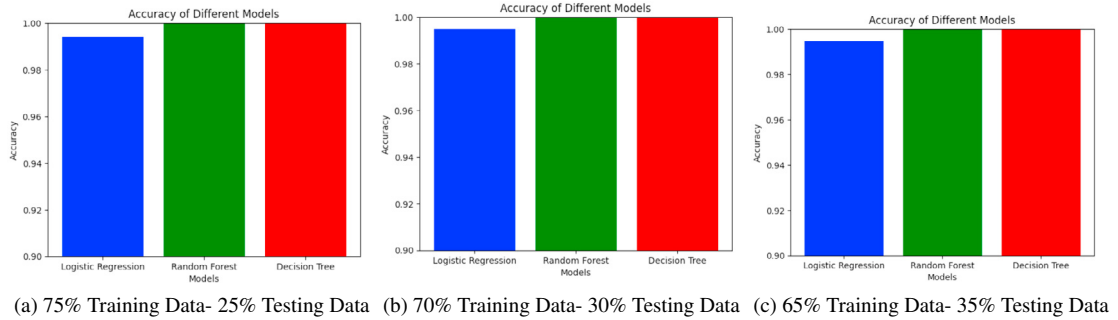


Fig. 8: Accuracy

Some of the rows in the dataset are abnormal distribution, hence, used scaling to normalize or standardize the data. To do that, Robust Scalar and MinMax Scalar [21] are used. (Note: Scaling was done after dividing the dataset into Train and Test Datasets).

## 5. Classification Models and Result Analysis

The classification algorithms namely Logistic Regression from Linear Model, Decision Tree classifier from Tree, and Random Forest Classifier from Ensemble methods are used.

Logistic regression despite its name, it is actually used for Classification and widely used for binary and multi-classifications. The basic equation for linear regression for multiple independent variables is  $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots \beta_i X_i$  [22]. Where as for logistic regression, probability log that an instance belongs to a certain class is used.  $P$  gives the probability that the instance belongs to the positive class (*class1*), and  $1 - P$  to the negative class (*class0*). The equation for binary classification is  $\ln(P/1 - P) = \alpha_0 + \alpha_1 X_1 + \alpha_2 X_2 + \dots \alpha_i X_i$  [22].

Decision Tree, a classification system based on machine learning algorithms divides a population into branch like segments, resulting in an inverted tree with leaf, internal, and root nodes. Because it is non-parametric, it can handle large and complicated datasets well without having a complicated parametric structure. Several decision trees are available, including CHAID, C4.5, CART, and QUEST [23].

Random Forest is a potent ensemble learning technique that increases the precision and resilience of machine learning models. It is well known for its ability to handle complicated datasets and high-dimensional feature spaces and is frequently used for both classification and regression problems. By training numerous trees on arbitrary selections of training data and characteristics, Random Forest expands on the idea of Decision Trees [24].

### 5.1. Results

To evaluate the performance of the above said classification algorithms generated confusion matrix for the data set. From confusion matrix metrics such as Accuracy, Recall, precision, and F1-score. This dataset has multiple outcomes, hence, used the confusion matrix for Multi-class Classification.

After following the steps in Figure 1 and apply the algorithms namely Logistic Regression, Decision Tree, Random forest. The results are shown in Figure 2, Figure 3, and Figure 4 in the form of confusion matrix. This confusion matrix is used to calculate precision, recall, and F1 score.

#### 5.1.1. Performance Metrics

Three ML models were used to detect DDoS attacks and there are four possible outcomes namely accuracy, precision, recall, and F1 Score for every possible outcome of applied model.

Figure 5, Figure 6, and Figure 7 gives the comparison of precision, recall, and F1 Score among three ML models by considering different ratio of training & testing data such as 75-25, 70-30, 65-35. By overall the precision of all the algorithms are shown in Figure 8 with different ratio of training & testing data. The accuracy of random forest, decision tree is 99.9% and logistic regression accuracy is 99.4%.

The results shows both decision tree and logistic regression are giving same accuracy because the dataset size is limited and manageable. Hence, decision tree is sufficient for the given data set. The random forest will work so accurate and handle large and unmanageable datasets with the help of ensemble process.

## 6. Conclusion and Future Work

In this paper, DDoS attack detection in SDN-IoT networks on dataset CSE-CIC-IDS2018 [19] using machine learning algorithms such as Random Forest, Decision Tree, and Logistic regression. Using the intrusion dataset CSE-CIC-IDS2018 and using various Machine Learning Models the results suggest that Random Forest and Decision Tree has best accuracy (99.9%) by considering both precision and recall both Random Forest and Decision tree both can be used for detecting DDoS Attacks in SDN IoT Networks.

But high accuracy does not mean that model is working better on the intrusion data. Due to this, in future, we will be using other datasets in large size for effective detection of DDoS attacks. Trying other Machine Learning Models on other datasets will help to build better Models

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